Aravindh Varadharaj

Full-Stack Developer · Three Python + Three Go Platforms · Next.js & React

Tiruppur | aravindhvaradha@gmail.com
GitHub | LinkedIn

SUMMARY

Builder at Amazon since 2021 in digital marketing, where I developed and automated internal workflows. I now build six distinct platforms: Python systems for enterprise open learning, historical famine analysis and animal welfare; and Go systems for general disaster impact, nuclear civil safety and wildlife-aware wireless efficiency. Each has a live React interface, tested API, explicit uncertainty and a documented production path. I am eager to build useful technology with communities that care about everyday impact.

SKILLS

Frontend engineering: Next.js 16, React 19, TypeScript 7, Tailwind CSS 4, server rendering, accessible responsive UI
Go systems: net/http, concurrency, streaming, idempotency, event sourcing, CRDTs, reconciliation loops
Python & intelligence: FastAPI, Pydantic, NumPy, pandas, forecasting, anomaly detection, explainability
Data & platform: Postgres/PostGIS design, Redis design, OpenTelemetry, Docker, GitHub Actions, versioned APIs
Safety & reliability: human approval gates, privacy by design, graceful degradation, property tests, race detection, model limitations

SELECTED PROJECTS

OpenSchool Enterprise — A complete free school platform for families and learning communities, including offline. · Python

Stack: Next.js 16, React 19, TypeScript 7, Python, FastAPI

- A governed Python/FastAPI platform creates mastery-aware plans, balanced curriculum coverage and printable offline guides for parents, teachers and administrators.
- Organization, role and curriculum contracts with auditable content-review state

FamineLens — Study historical evidence carefully—and connect its lessons to accountable early warning. · Python

Stack: Next.js 16, React 19, TypeScript 7, Python, FastAPI

- A sourced comparison of the Great Indian Famine (1876–1878) and Bengal Famine (1943) pairs contested historical evidence with an uncertainty-aware modern scenario model.
- Historical claims retain official/reference sources and disputed interpretations

Kindred — Connect animal welfare reports to timely, humane and accountable care. · Python

Stack: Next.js 16, React 19, TypeScript 7, Python, FastAPI

- A privacy-aware Python network maps coarse wildlife reports, shelter capacity and environmental pressure, then builds a transparent human-review queue.
- Wildlife locations are coarsened; every priority remains an expert-review recommendation

TerraZero — Turn public hazard signals into understandable impact and preparedness scenarios. · Go

Stack: Next.js 16, React 19, TypeScript 7, Go, net/http

- A Go multi-hazard twin combines live public USGS earthquakes with transparent earthquake, volcano, tsunami, flood and wildfire impact-zone scenarios.
- Fixed public USGS endpoint with bounded fetch, cache and fallback behavior

Nuclear Legacy & Civil Safety — Learn from civilian nuclear accidents without turning uncertainty into fear or false certainty. · Go

Stack: Next.js 16, React 19, TypeScript 7, Go, net/http

- A Go civil-safety observatory compares Three Mile Island, Chernobyl and Fukushima and visualizes educational monitoring and recovery scenarios.
- Strictly excludes weapon design, targeting and attack modelling

EcoSignal — Reduce unnecessary wireless transmission while preserving essential connectivity. · Go

Stack: Next.js 16, React 19, TypeScript 7, Go, net/http

- Enumerates configurable access-point/gateway power plans under a minimum coverage constraint
- Compares current and optimized RF activity maps, device duty cycles and energy while preserving an essential emergency gateway

EXPERIENCE

Sr. Digital Marketing Associate — Amazon 2021 – Present

- Developed and automated internal tools and workflows for my team, replacing repetitive manual tasks with reliable, reusable solutions.
- Worked with stakeholders to identify process gaps, then designed, built, and supported automation that the team adopted in daily work.
- Grew from a marketing role into hands-on development, building a portfolio of full-stack Python projects outside my core responsibilities.

EDUCATION

Bachelor of Engineering (B.E.) in Mechanical Engineering